CAMILA FARIAS AMORIM, M.S., Ph.D.

Immunology, Translational Medicine, Computational Biology and Data Sciences

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PROFESSIONAL SUMMARY

Clinical Immunology Ph.D. scientist with over 12 years of experience in biomarker/target discovery for early diagnostics, immunotherapy, and vaccines, assay and product development. Extensive understanding of multi-tissue biology, pathogenesis, cellular and molecular assays, and transcriptional profiling from Omics large data on preclinical/clinical research. An established computational biologist with expertise in clinical operations, statistical, and data analysis. Mined and integrated multi-omics multi-modal NGS data using R, Bioconductor, statistical inference, linear modeling, and machine learning methods (AI/ML). Taught bioinformatics programming and Next-generation sequencing (NGS) at UPenn for 5 years. Coordinated Biomarker and Molecular Assay Development strategies from sample collection, data analysis to application. Highly communicative within multi-disciplinary, cross-functional teams. Established scientist with a proven track record of independence, oral & written communication skills (18+ publications, oral and poster presentation awards).

I am committed to utilizing data science to enhance patient care and quality of life.

PROFESSIONAL EXPERIENCE

Century Therapeutics 2024-current

Role: Senior Scientist Computational Biology and Data Sciences

- Led que Preclinical Research NGS Molecular Profiling and Biomarker discovery program and coordinated Cellular/Molecular Assay Development (qPCR, TaqMan, Flow Cytometry, CITE-seq) strategies from study design, sample collection, data analysis to application for actionable improvement of CAR cell products.
- Organized and led the Omics meeting (>20 wet/dry-lab scientists) to discuss interpret and apply NGS-related insights to company-wide programs.
- Collaborated with scientists from Research Development: cell engineering, iPSC biology, Immunology, Cell Differentiation, as well as Early Clinical and Translational Development to design experiments and analyze data, ensuring statistically valid data collection, sampling methods, and data interpretation – across company branch sites (NJ, Philadelphia, Boston).
- Analyzed in-house and publicly available large-scale NGS, Flow cytometry, clinical and pre-clinical research datasets to support the development of universal iPSC-derived allogeneic cell products targeting hematological, solid tumors, and auto-immune diseases.
- Built and maintained data analysis pipelines, integrating diverse biological data sources and ensuring reproducibility to drive product improvements by computational programming.
- Communicated complex computational analyses to diverse audiences, including external collaborators, investors (all NGS-related data1 and data2) and cross-functional teams. Demonstrated entrepreneurial initiative and the ability to thrive in a dynamic, fast-pacing, Biotech-startup environment.

University of Pennsylvania, Philadelphia, Pennsylvania 2021-2024

Role: Research Associate

- Orchestrated international projects, including 5-year, >$15M NIH RO1 grants (Brazil-USA).
- Reported to 4 Principal Investigators, contributing to grant and study reports, and final publications.
- Oversaw and executed all aspects of early-stage clinical development research: engagement with patients at the hospitals (South America); patient recruitment; collected and processed clinical samples for immune-assays and NGS; organization of clinical inventory, data and protocol development, built patient database; analyzed output data; interpreted qualitative and quantitative findings; and developed biomarker program-based strategies.
- Led an assay development project for early identification of patients with poor response to standard therapy: qPCR rapid assays targeting gene expression (GNLY, IL1B) and skin parasite and bacterial burdens. Project in direct collaboration with physicians and hospitals in South America.
- Established the functional role of cytotoxic CD8+ T cells in human cutaneous leishmaniasis immunopathology and delayed response to therapy based on gene expression RNA-seq profiling.
- Matrix team leader on pre-clinical studies identifying and validating gene-therapy (IL-1a, IL-1b, IL-1R) and microbiome-therapy (*Staphylococcus aureus*) targets for *Leishmania braziliensis* skin infection.
- Collaborated with bench scientists (*in vivo*/in vitro) and bioinformaticians (KOLs) in projects to (i) dissect the role of hypoxia in influencing the microenvironment, CD8+ T cell and Neutrophil functions in skin inflammatory diseases; (ii) describe the role of CD8+ T cells expressing NKG2D in causing immunopathogenesis in skin; (iii) biomarker discover of CD8+ T cell-related effector functions targeted for successful immunotherapy (Perforin-inhibitor; cytolysis-inhibitor Tofacitinib, NLRP3-inhibitors Glyburide/MCC950).
- Hands-on analyzed and integrated multi-omics datasets (mouse and human): RNAseq, Dual RNA-seq, single-cell RNAseq, CITE-Seq, 16S-seq (microbiome), WGS, proteomics, qPCR, and clinical human metadata. Expert in multivariate linear modeling and supervised classification AI/ML methods.

- Performed computational transcriptional (RNA-seq) and microbiome (16S-seq) profiling between skin chronic diseases: psoriasis, atopic dermatitis, and cutaneous leishmaniasis, identifying distinct inflammatory and bacterial patterns.
- Worked on a mRNA vaccine project for cutaneous leishmaniasis (based on enhancing skin T resident memory effector cell functions). Analyzing and managing multiple single-cell CITE-seq datasets for immuno-genetics profiling.
- Direct contributor on a collaborative preclinical/clinical project evaluating the role of Foxp3 RORYt T regulatory cells in shifting the IL-17 vs. IFN-y inflammatory pathways in the skin co-infection between *L. braziliensis* and *S. aureus*.
- Subject matter expert in Clinical Research, Computational Biology, transcriptional profiling, and NGS in a Basic Immunology group of experimental scientists (mice). Established and managed the NGS laboratory protocols and dataset database in high-performance computing (HPC)/server clusters.
- Organized and led departmental (>60 participants) and internal group meetings for project progress reports.
- Supervised, mentored, and guided laboratory technicians (3), graduate students (2), postdocs (4), and senior project staff (1) on cellular and molecular wet-lab experiments desired for early-to-late drug/biomarker discovery, laboratory troubleshooting, and protocol adaptation, translational/validation studies from clinical and experimental samples.

University of Pennsylvania, Philadelphia, Pennsylvania 2017-2021
Role: Postdoctoral Researcher

- Conducted data mining focused on patient stratification to strategize precision medicine hypotheses (E.g. based on gene expression, skin microbiomes, hypoxia levels).
- Analyzed multiple clinical skin and blood bulk RNA-seq datasets of patients infected by *L. braziliensis*, pinpointing CD8+ T cell and NLRP3 inflammasome-driven IL1B upregulation as the key inflammatory components in cutaneous leishmaniasis.
- Developed a R statistical computational algorithm based on linear models and patient gene variability concepts to predict early therapy failure biomarkers: GNLY, GZMB, PRF1, and IL1B.
- Proficient in estimating immune-cell states and functions from unstructured bulk RNA-seq data, including Cytotoxic T CD8+ cells, NK cells, Neutrophils, Monocytes, Fibroblasts, and Keratinocytes.
- Created interactive data visualizations (ShinyApp) for easy data analyses and interpretation by non-coding scientists.
- Analyzed a blood RNA-seq dataset, identifying a primed systemic monocyte gene signature activated by IFN-γ in patients with localized cutaneous leishmaniasis.
- Developed predictive computational pipelines for NGS data, discovering the CCL3/4 and CCR5 chemotaxis axis's role in recruiting pathogenic CD8+ T cells. Coordinated a project to successfully block CCR5 therapeutically (Maraviroc – FDA approved), abrogate skin inflammation, and heal disease in mice.
- Fostered collaborations leading to insights into CD8+ T cell immunosenescence in leishmaniasis and the development of a selective whole genome amplification (SWGA) sequencing method for parasite transcript analysis.
- Presented numerous talks and posters at yearly international conferences such as Immunology (AAI), RStudio, and Bioconductor.

Center of Host-Microbiome Interactions (CHMI) - NGS Core, University of Pennsylvania 2017-2023
Role: Teaching Assistant and NGS consultant

- Collaborated across UPenn labs in bioinformatics, cancer, immunology, and infectious diseases, serving as an academic CRO for big data analysis, NGS processing, interpretation, and project goal guidance.
- Instructed >100 graduate students, postdoctoral fellows, and faculty staff per semester in R-based RNAseq workflows, data science best practices, data visualization, biotechnology, and transcriptomics biological interpretation at DIYtranscriptomics.com course (graduate curriculum).
- Developed scalable computational workflows, emphasizing reproducible coding practices, and contributed to open-source NGS genomics algorithms (code capsules).
- Mentored high school students from underrepresented backgrounds and underserved schools in Pennsylvania on scientific research and biology through the UPenn LENS program.
- Published a mBio manuscript introducing bioinformatics workflows for transcriptomic analysis using publicly available RNAseq datasets.

Federal University of Bahia, Bahia, Brazil 2010-2017
Masters and Ph.D. graduate in Immunology Health Sciences

- Regularly engaged with patients from endemic areas, addressing viral and parasitic infections (HTLV-1, *Leishmania braziliensis*), cancer (Adult T-cell leukemia/lymphoma), and autoimmune and neurological diseases (psoriasis, rheumatoid arthritis, HTLV-1-Associated Myelopathy/Tropical Spastic Paraparesis).
- Led development programs, research design, subject recruitment, and the collection of blood and skin samples.
- Contributed to pre-clinical studies and clinical trials in Brazil's infectious disease endemic regions.
- Collaborated on in vitro anti-parasitic drug efficacy projects, identifying IQG-607 as effective against *Leishmania spp.* and DI4G as a potent IFN-y and CD8+ T cell immunomodulator.

- Phenotypically and functionally characterized innate cells (CD56+ CD16-/+ NK, CD14+/- CD16+/- monocytes/monocyte-derived macrophages) in HTLV-1-infected patients using flow cytometry and in vitro assays.

HIGHLIGHTED SKILLS AND KEYWORDS

Immunology

DNA/RNA extraction; qRT-PCR; high parametric multi-color flow cytometry/FACS; ELISAs; Cell culture; *in vitro* immune-pathway blockade with monoclonal antibodies; IHC/H&E; PBMC purification from blood (Ficoll)

Bioinformatics

R; python; C/C++; bash; AWS; Unix/Linux/HPC/OS; command-line tools; shell scripting; RNAseq; Dual-RNAseq; single-cell RNAseq; CITE-Seq; 16S-seq; WGS; tidyverse; Seurat; CellRanger; ggplot; plotly; LaTex; Read mapping; Bioconductor; AI/ML; PCA; classification/stratification; hierarchical clustering; dimensionality reduction; pattern recognition; pathway analysis; linear/non-linear regression models; GSEA; ontology; GEO NCBI; FastQC; DESeq2; SAMtools; correlation networks; Git; version control; cloud; ShinyApp; RMarkdown; Docker; CodeOcean capsule

Personal

Expert in interacting with translational multi-disciplinary teams; enthusiastic colleague; efficient verbal and written communicator; data-driven scientist; excellent in generating high-quality detailed or summarized reports and talks; efficient with timeline execution; independent learner of novel computational tools.

EDUCATION

2014-2017	**Doctor of Immunology, Ph.D., Clinical Health Sciences – UFBA, Brazil** Thesis: The role of NK cells in the Human T lymphotropic virus (HTLV-1) carriers
2012-2013	**Master of Immunology, M.S., Clinical Health Sciences – UFBA, Brazil** Thesis: Functional activity of monocytes and macrophages in HTLV-1 infected subjects
2008-2011	**Bachelor of Biomedicine, Clinical Health Sciences** Bahia School of Medicine and Public Health, Salvador, Bahia, Brazil

PROFESSIONAL AND OTHER LEADERSHIP ROLES

Member of professional societies

2020-current American Society of Immunologists (AAI)

Reviewer

2021-current Frontiers in Cellular and Infection Microbiology (2022-); The Journal of Infectious Diseases (2022-); JID Innovations (2021-)

AWARDS AND ACHIEVEMENTS

2023	**Recognition of Scientific Excellence - American Society of Tropical Medicine and Hygiene (ASTMH)**, ASTMH 2023 Conference, Chicago, USA
2023	**American Association of Immunologists (AAI) Oral Presentation Award - First Place**, Woods Hole Immunoparasitology Group, Marine Biological Laboratory, University of Chicago
2023	**WHIP 2023 Travel Award**, from National Institutes of Health (NIH), Woods Hole Immunoparasitology Group, Marine Biological Laboratory, University of Chicago
2022	**WHIP 2022 Travel Award**, from National Institutes of Health (NIH), Woods Hole Immunoparasitology Group, Marine Biological Laboratory, University of Chicago
2017	**Ph.D. fellow scholarship award** for developing a scientific research project abroad - by CAPES, Brazil

SELECTED PUBLICATIONS

Citations on Google Scholar 12/28/2023 = 251
More publications and citation numbers at Google Scholar Profile and Web of Science
(*Indicates equal contribution)

1. E. A. Fowler, **C. F. Amorim**, K. Mostacada, A. Yan, L. A. Sacramento, R. A. Stanco, E. D. S. Hales, A. Varkey, W. Zong, G. D. Wu, C. I. de Oliveira, P. L. Collins, F. O. Novais. Neutrophil-mediated hypoxia drives pathogenic CD8+ T cell responses in cutaneous leishmaniasis. Journal of Infectious Diseases. (2024), doi: 10.1172/JCI177992.
2. L. A. Sacramento, **C. F. Amorim**, C. G. Lombana, D. Beiting, F. Novais, L. P. Carvalho, E. M. Carvalho, P. Scott, CCR5 promotes the migration of CD8+ T cells to the leishmanial lesions. PLoS Pathogens (2024), doi: 10.1371/journal.ppat.1012211.

3. **C. Farias Amorim**, V. M. Lovins, T. P. Singh, F. O. Novais, J. C. Harris, A. S. Lago, L. P. Carvalho, E. M. Carvalho, D. P. Beiting, P. Scott, E. A. Grice, Multiomic profiling of cutaneous leishmaniasis infections reveals microbiota-driven mechanisms underlying disease severity. Sci. Transl. Med. 15, eadh1469 (2023).
4. O. A. Pilling, C. A. Grace, J. L. Reis-Cunha, A. S. Berry, M. W. Mitchell, J. A. Yu, C. Malekshahi, E. Krespan, C. K. Go, C. Lombana, Y. S. Song, **C. F. Amorim**, A. S. Lago, L. P. Carvalho, E. M. Carvalho, D. Brisson, P. Scott, D. C. Jeffares, D. P. Beiting, Selective whole-genome amplification reveals population genetics of Leishmania braziliensis directly from patient skin biopsies. PLoS Pathogens (2022), doi: 10.1371/journal.ppat.1011230.
5. L. A. Sacramento, **C. Farias Amorim**, T. M. Campos, M. Saldanha, S. Arruda, L. P. Carvalho, D. P. Beiting, E. M. Carvalho, F. O. Novais, P. Scott, NKG2D promotes CD8 T cell-mediated cytotoxicity and is associated with treatment failure in human cutaneous leishmaniasis. PLoS Negl. Trop. Dis. 17, e0011552 (2023).
6. T. P. Singh, **C. Farias Amorim**, V. M. Lovins, C. W. Bradley, L. P. Carvalho, E. M. Carvalho, E. A. Grice, P. Scott, Regulatory T cells control Staphylococcus aureus and disease severity of cutaneous leishmaniasis. J. Exp. Med. 220 (2023), doi:10.1084/jem.20230558.
7. **C. Farias Amorim**, F. O Novais, B. T. Nguyen, M. T. Nascimento, J. Lago, A. S. Lago, L. P. Carvalho, D. P. Beiting, P. Scott, Localized skin inflammation during cutaneous leishmaniasis drives a chronic, systemic IFN-γ signature. PLoS Negl. Trop. Dis. 15, e0009321 (2021).
8. A. S. F. Berry, **C. Farias Amorim**, C. L. Berry, C. M. Syrett, E. D. English, D. P. Beiting, An Open-Source Toolkit To Expand Bioinformatics Training in Infectious Diseases. MBio. 12, e0121421 (2021).
9. C. H. Fantecelle, L. P. Covre, R. Garcia de Moura, H. L. de M. Guedes, **C. F. Amorim**, P. Scott, D. Mosser, A. Falqueto, A. N. Akbar, D. C. O. Gomes, Transcriptomic landscape of skin lesions in cutaneous leishmaniasis reveals a strong CD8+ T cell immunosenescence signature linked to immunopathology. Immunology. 164, 754–765 (2021).
10. F. O. Novais*, **C. F. Amorim***, P. Scott, Host-Directed Therapies for Cutaneous Leishmaniasis. Front. Immunol. 12, 660183 (2021).
11. M. T. Nascimento, M. Franca, A. M. Carvalho, **C. F. Amorim**, F. Peixoto, D. Beiting, P. Scott, E. M. Carvalho, L. P. Carvalho, Inhibition of gamma-secretase activity without interfering in Notch signalling decreases inflammatory response in patients with cutaneous leishmaniasis. Emerg. Microbes Infect. 10, 1219–1226 (2021).
12. A. S. F. Berry, K. Johnson, R. Martins, M. C. Sullivan, **C. Farias Amorim**, A. Putre, A. Scott, S. Wang, B. Lindsay, R. N. Baldassano, T. J. Nolan, D. P. Beiting, Natural Infection with Giardia Is Associated with Altered Community Structure of the Human and Canine Gut Microbiome. mSphere. 5 (2020), doi:10.1128/mSphere.00670-20.
13. **C. F. Amorim**, F. O. Novais, B. T. Nguyen, A. M. Misic, L. P. Carvalho, E. M. Carvalho, D. P. Beiting, P. Scott, Variable gene expression and parasite load predict treatment outcome in cutaneous leishmaniasis. Sci. Transl. Med. 11 (2019), doi:10.1126/scitranslmed.aax4204.
14. **C. F. Amorim**, N. B. Carvalho, J. A. Neto, S. B. Santos, M. F. R. Grassi, L. P. Carvalho, E. M. Carvalho, The Role of NK Cells in the Control of Viral Infection in HTLV-1 Carriers. J. Immunol. Res. 2019, 6574828 (2019).
15. **C. F. Amorim**, A. S. Souza, A. G. Diniz, N. B. Carvalho, S. B. Santos, E. M. Carvalho, Functional activity of monocytes and macrophages in HTLV-1 infected subjects. PLoS Negl. Trop. Dis. 8, e3399 (2014).
16. M. Guerra, T. Luna, A. Souza, **C. Amorim**, N. B. Carvalho, L. Carvalho, D. Tanajura, L. S. Cardoso, E. M. Carvalho, S. Santos, Local and systemic production of proinflammatory chemokines in the pathogenesis of HAM/TSP. Cell. Immunol. 334, 70–77 (2018).
17. **C. F. Amorim**, L. Galina, N. B. Carvalho, N. D. M. Sperotto, K. Pissinate, P. Machado, M. M. Campos, L. A. Basso, V. S. Rodrigues-Junior, E. M. Carvalho, D. S. Santos, Inhibitory activity of pentacyano(isoniazid)ferrate(II), IQG-607, against promastigotes and amastigotes forms of Leishmania braziliensis. PLoS ONE. 12, e0190294 (2017).
18. N. B. Carvalho, F. V. de Oliveira Prates, R. de C. da Silva, M. E. F. Dourado, **C. F. Amorim**, P. R. L. Machado, F. G. Pacheco, T. W. F. Corte, P. Machado, D. S. Santos, E. M. de Carvalho, In Vitro Immunomodulatory Activity of a Transition-State Analog Inhibitor of Human Purine Nucleoside Phosphorylase in Cutaneous Leishmaniasis. J. Immunol. Res. 2017, 3062892 (2017).
19. A. M. Carvalho, **C. F. Amorim**, J. L. S. Barbosa, A. S. Lago, E. M. Carvalho, Age modifies the immunologic response and clinical presentation of American tegumentary leishmaniasis. Am. J. Trop. Med. Hyg. 92, 1173–1177 (2015).
20. A. S. Souza, **C. F. Amorim**, N. Carvalho, S. B. Santos, E. M. Carvalho, Impairment of humoral and cellular immune response to tetanus toxoid in HTLV-1 infected individuals. Retrovirology. 11, P66 (2014).

SELECTED CONFERENCE PARTICIPATION .

2023	Woods Hole Immunoparasitology meeting, talk: “Multi-omic profiling of cutaneous leishmaniasis infections reveals microbiota-driven mechanisms underlying disease severity”, Woods Hole, MA
2023	American Society of Tropical Medicine and Hygiene 2023 Annual Meeting at the Hyatt Regency Chicago, talk: “Multiomic profiling of cutaneous leishmaniasis infections reveals microbiota-driven mechanisms underlying disease severity”, Chicago IL, USA
2022	Woods Hole Immunoparasitology meeting, talk: “The skin microbiome enhances transcriptional inflammatory signatures and delays clinical resolution in cutaneous leishmaniasis”, Woods Hole, MA
2022	Immunology2022 Annual Meeting of The American Association of Immunologists, Portland, OR
2021	Woods Hole Immunoparasitology meeting, poster: “*Staphylococcus aureus* contributes to cutaneous leishmaniasis pathology by stimulating neutrophil skin infiltration and enhancing pro-inflammatory gene expression”, Woods Hole, MA

2021 Immunology2021 Annual Meeting of The American Association of Immunologists, Virtual

SELECTED COURSES .

2022 Probability and Statistical Inference - The 27th Summer Institute in Statistical Genetics, Biostatistics School of Public Health, University of Washington, Seattle, WA

2022 Infectious Diseases, Immunology and Within-Host Models - The 14th Summer Institute in Modeling for Infectious Diseases, Biostatistics School of Public Health, University of Washington, Seattle, WA